

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: T.Y.B. Tech. Branch : Mech. Engg. Semester :VI Subject Code & Name: BTMPE604D Robotics Max Marks: 60 Date: 19/07/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What is Industrial Automation ? Discuss various types of Industrial automation	U/CO1	6
B)	Explain various robot configurations with sketches.	U/CO1	6
C)	Write short note on Robotics control system	U/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Derive the forward kinematic equation for 2 DOF robot manipulator having link length L1, L2 & Orientation angle θ_1 , θ_2	E/CO2	6
B)	Write and explain Homogeneous transformation matrix for rotation and Translation.	U/CO2	6
C)	Write a short note on Trajectory planning in robotics.	U/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Distinguish between Hydraulic, Pneumatic and Electrical Drives.	E/CO3	6
B)	Describe construction and working of AC Motors	U/CO3	6
C)	Describe construction and working of Stepper Motors	U/CO3	6
Q.4	Solve Any Two of the following.		12
A)	Describe construction and working of LVDT to measure the linear displacement.	E/CO4	6
B)	Describe construction and working of Proximity Sensors	U/CO4	6
C)	Explain Magnetic gripper in details	U/CO4	6
Q. 5	Solve Any Two of the following.		12

A)	Explain advantages and disadvantages of lead through programming in details	U/CO5	6
B)	Discuss economic analysis of Robots	U/CO5	6
C)	Explain the role of robot in material handling	U/CO5	6
	*** End ***		

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